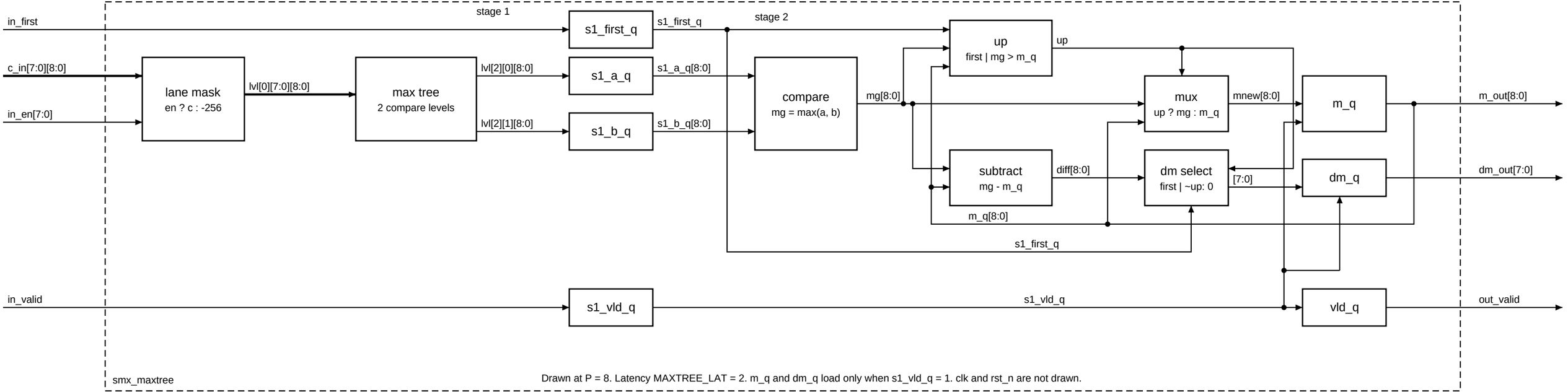
stage 1
stage 2
in_first
c_in[7:0][8:0]
in_en[7:0]
lane mask
en ? c : -256
lvl[0][7:0][8:0]
max tree
2 compare levels
lvl[2][0][8:0]
lvl[2][1][8:0]
s1_first_q
s1_a_q
s1_b_q
s1_first_q
s1_a_q[8:0]
s1_b_q[8:0]
compare
mg = max(a, b)
mg[8:0]
up
first | mg > m_q
up
mux
up ? mg : m_q
mnew[8:0]
m_q
m_out[8:0]
subtract
mg - m_q
diff[8:0]
dm select
first | ~up: 0
[7:0]
dm_q
dm_out[7:0]
m_q[8:0]
s1_first_q
in_valid
s1_vld_q
s1_vld_q
vld_q
out_valid
smx_maxtree
Drawn at P = 8. Latency MAXTREE_LAT = 2. m_q and dm_q load only when s1_vld_q = 1. clk and rst_n are not drawn.